[Experiment, Analysis and Benchmark] How Much Entity Integrity Matters

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***Abstract*—Entity integrity is the principle that every entity of an application domain ought to be represented uniquely within a database. This is operationalized by using candidate keys, including the primary key. In practice, candidate keys are turned off to ease data acquisition, or the same entity is assigned multiple identifiers. Each of these malpractices can result in entity integrity faults. Despite its fundamental importance and despite the broad agreement that high quality analytics requires high quality data, we are unaware of studies that quantify the impact of entity integrity faults on analytical tasks. For this reason, we propose a methodology that enterprises can use to systematically quantify the impact of entity integrity faults on database queries to guide decision making. The methodology explores different dimensions of data quality, such as uniqueness and completeness, but also metrics that quantify the impact of violations on the precision, recall and performance of database queries. As a proof of concept, we apply the framework to the TPC-H benchmark. This illustrates how much entity integrity matters.**

***Index Terms*—Entity Integrity, Key, Query Accuracy, Error Propagation**

I. Introduction

Data is the new oil[1]. This quote is one example how much importance organizations attribute to data in the 21st century, in an attempt to implement data-driven decision making, extract actionable insight, and gain competitive advantage. However, the extent of these benefits is limited by the quality of the data [1], [2]. One of the most important principles which is necessary to guarantee high quality data is data integrity. In relational databases, entity integrity is operationalized by the use of primary keys. Here, each table has a single, non-null primary key to uniquely identify every record. This was already outlined by Codd when he proposed the relational model of data [3]. More generally, entities of an application domain are captured by records in a table, and candidate keys constitute minimal sets of table columns such that no entity has any non-null value on any column of the key, and no two different entities have values matching on every column of the key. The primary key of a table is a distinguished candidate key. Not using any candidate key means that tables may contain duplicate records, or miss information that makes it impossible to identify entities uniquely. This will result in database instances that misrepresent the current state of an application domain, which will produce faults in downstream tasks such as analytics. In terms of database queries and any other data-driven analytical methods such as AI, machine learning and statistics, results may be inaccurate, incomplete, or inconsistent, causing poor decision making, operational inefficiency, loss of trust and revenue, compliance risks, and waste of resources.

Despite its importance, entity integrity is not always guaranteed in practice which may be due to the nature of the underlying data, or even be on purpose to ease with the acquisition of data or improve operational efficiency [4]. Even when primary keys are being used, several issues can occur. Scenarios, such as a customer with multiple customer ids, are far from uncommon [5]. Reliance on surrogate keys (artificial identifiers) to serve as the primary key without the use of any natural candidate key (also called business keys) can lead to recording the same real-world entity multiple times, each time with a different artificial identifier. We refer to these records as semantic duplicates. The principle of entity integrity can even be abused, and situations where the same person has several social security numbers to claim benefits multiple times are possible. Furthermore, the sole focus on primary keys for entity integrity management poses challenges when integrating data from various sources across the same domain. For example, the artificial identifiers of products might not align across different datasets. Lastly, for big data applications that require horizontal scaling, facilitated through distributed systems, synchronizing the parallel generation of surrogate key values can become problematic [6].

Entity integrity can be initiated through robust schema design, and its efficient maintenance cab be facilitated through effective generation of artificial identifiers and safe updates of values on every business key. This requires continuous monitoring in a dynamic, ever evolving environment that will challenge database practitioners. Due to their role in implementing entity integrity, keys are essential for all data models. They are fundamental in many classical and modern areas of data management. Knowledge about keys enables us to (i) uniquely reference entities across data repositories, (ii) minimize data redundancy at schema design time to process updates efficiently at run time, (iii) provide better selectivity estimates in cost-based query optimization, (iv)

[1]https://sheffield.ac.uk/cs/people/academic-visitors/clive-humby

TABLE I: Tables with Entity Integrity Problems

(a) Table CITY

zip	city
1001	Shelbyville
1001	Shelbyville
NULL	Springfield

(b) Table CUSTOMER

cust_id	name	address	zip
1497	Homer	742 Green Rd	555
1824	Homer	742 Green Rd	555
1962	Montgomery B	1000 Ma Lane	1001

provide a query optimizer with new access paths that can lead to substantial speed-ups in query processing, (v) allow the database administrator (DBA) to improve the efficiency of data access via physical design techniques such as data partitioning or the creation of indexes and materialized views, and (vi) provide new insights into application data. Modern applications raise the importance of keys further.

General knowledge stipulates that if entity integrity is not enforced in data repositories the accuracy of queries can suffer as a result. Yet, in spite of all of this, to our surprise, there is no methodology that quantifies the effects of entity integrity violations. In particular, to the best of our knowledge, no prior work has focused on how the accuracy of anticipated query workloads is impacted by entity integrity faults. This is particularly surprising as thought leaders in data quality research have called for work that quantifies the impact of data quality problems [7].

In view of the fundamental importance of entity integrity and very common malpractices, we establish a methodology that will enable enterprises to quantify the impact of entity integrity faults on database workloads. These insights can then be used to steer decision making regarding entity integrity enforcement. A proof of concept will be provided by applying the framework to the well-known relational TPC-H benchmark.

We investigate to which degree query accuracy of the 22 benchmark queries reduces when introducing either of the following entity integrity violations. First, we will investigate the effect that the duplication of records has. An example of such a violation is provided in Table Ia with primary key *zip* not being enforced: the city record with *zip* 1001 is repeated. Asking for all records within a given *zip* range may return duplicate records, and using `distinct` is slow due to a `unique` index not being available as the primary key is not enforced. Secondly, we analyse the impact of missing values on primary key attributes. An example can also be seen in Table Ia where the city record relating to 'Springfield' carries no value on *zip*. Asking for the name of cities within a given *zip* range that features the actual *zip* value of 'Springfield' will not return 'Springfield'. Finally, we analyse how queries are affected when primary keys are enforced, yet entity integrity is violated [5]. We refer to such scenarios as deep entity integrity violations and an example can be seen in Table Ib. Here, the customer 'Homer' refers to the same entity but has been recorded as different customers since their primary key values on *cust_id* are different. Here, the composite business key {*name*,*address*} is not enforced, making it possible to assign different customer ids to the same customer. Attempting to retrieve all orders from the customer with *cust_id* '1497' will only give partial results in case orders from 'Homer' at '742 Green Rd' are meant. We note that in general not all non-key attributes have to match exactly on different representations of the same entity [5]. Enforcement of meaningful keys can help to prevent such situations.

Contributions: As these examples represent common integrity pitfalls, it is our aim to quantify how they impact the accuracy of queries. The contributions of our research can be summarized as follows.

- We establish a methodology for quantifying the impact of entity integrity faults on database workloads.
- The extent of faults will be measured in terms of the uniqueness and completeness requirements for primary keys, but also by the violation of alternate business keys.
- The impact on database workloads is measured by precision, recall, and efficiency of operations.
- It is shown how the size of data affects the impact.
- The effectiveness of the methodology is analysed experimentally on the 22 queries of TPC-H benchmark.
- Decisions around entity integrity management are presented based on applying our framework.

Our results emphasize the importance of using database keys for maintaining entity integrity, in particular the use of primary and alternate business keys. Enterprises are encouraged to apply our methodology to their databases, which will quantify the impact any entity integrity faults have on any of their database operations. The ability to quantify the impact can determine the level of investment in data governance to maximise the likelihood that expectations in high quality analytical outcomes can be met.

Organization: We discuss related work in Section II. Afterwards, we present our methodology in Section III. In Section IV, as a proof of concept, a comprehensive experimental analysis showcases how our methodology is applied, and what insights it can reveal. Finally, we conclude our work in Section V and discuss future research.

II. RELATED WORK

We position our contribution as a motivator for more research and development into data governance tools that facilitate high quality analytical outcomes and data-driven decision making. Likewise, organizations can apply our methodology to monitor how the entity integrity of their data affects the quality of their operations, and address concerns accordingly. To understand how our work is positioned within the rich literature, we will now review related work on data integrity in particular, data quality in general, as well as database repairs and consistent query answering.

A. Data Integrity

Since the very beginning of Codd's relational model of data [3], data integrity has always been a cornerstone for enabling fundamental data management tasks. While research has surfaced more than 100 classes of data integrity constraints [8], there are only three classes that enjoy native support within database systems. These comprise domain

constraints, key constraints, and foreign key constraints, addressing Codd's three fundamental data integrity rules: domain integrity, entity integrity, and referential integrity [9]. Entity integrity is based on the principle that every entity of the application domain is represented uniquely within the database system. Traditionally, the principle is enforced by the concept of a database key, which is a set of attributes such that no two different records must exist in a database instance that have values matching on all the attributes of the key [9], [10]. Primary keys are distinguished database keys that provide us with a `unique` index, ensuring that records can be accessed quickly using the values of the record on the primary key. No column of a primary key must feature any null marker occurrence, which ensures that every entity can be accessed efficiently [9]. Due to their role in implementing entity integrity, keys are essential for all data models, including relational [11], conceptual [12], object-relational [13], XML [14], RDF [15], temporal [16], spatial [17], probabilistic [18], and graph [19] models. Application of database keys encompass traditional data management areas such as data modelling, database design, indexing, and query optimization, but extend to modern tasks such as data integration [20], duplicate detection [21] and entity resolution [22], [23], database repairing and consistent query answering [24], [25], data cleaning [26], [27], data profiling [28], and the reliable use of large language models and AI agents for data processing [29]. Most existing work focuses on methods to enforce data integrity or detect them.

While undisputed that a lack of entity integrity may distort the output of queries there is a need to quantify, through experiments, how different types of entity integrity violations affect the results of various types of queries [1], [30]. To conduct such comprehensive studies we first need to outline a methodology that guides this process. Applying this methodology can then be used to reveal which violations are most inhibiting to specific application workloads.

B. Data Quality

In our work, we investigate different types of entity integrity faults, comprising the duplication of records, missing values on primary key columns, and violations of business keys. Interestingly, these types of violations form the very essence of important data quality dimensions, namely uniqueness, completeness, and consistency, respectively [31]–[33]. In fact, quantifying the impact of entity integrity faults on database workloads on a regular basis will also motivate support towards upholding the important currency and timeliness dimensions of data as well. Frameworks for assessing the quality of data evolve regularly [34]–[36], demonstrating how important and challenging advances in improving data quality are.

The very reason data quality research has introduced different dimensions and various measures for quantifying the quality of data with respect to these dimensions is to help organizations answer the question whether their data is fit for purpose [37]. The methodology we propose aims at quantifying how fit the data is for processing data workloads in terms of their level of entity integrity.

Any new analytical movement, such as data warehousing [38], big data [39], data science [40], [41] or data-centric AI [42]–[44] comes back to the garbage in, garbage out principle: any insight brought forward by analytics can only ever be as good as the quality of the data the analytics is applied to. Our methodology will make it possible to quantify how the quality in terms of entity integrity affects the quality of database queries.

C. Repairs and Consistent Query Answering

The area of consistent query answering has originated from the very observation that data integrity faults cause incorrect answers in queries [24], [25]. The basic idea is to return only those answers that are certain to occur in every repair of the integrity faults. A repair of inconsistent data emerges from minimal changes to the data that restore consistency. The definition depends on what types of operations are permitted to change the data, such as insertions, deletions or modifications of data values or records [45]. The approach makes it possible to execute database workloads with confidence despite inconsistent data [46], which becomes necessary when it is unclear which repair is the right one. In this sense, consistent answers miss all of those query answers that would be returned on the right repair but not on other repairs. From this point of view, our methodology provides an analysis of what query answers could be missing when only certain ones are available. Note that the area of database repairs and consistent query answering has seen a prolific trajectory in terms of theory [47], [48], in particular recent work on primary and foreign keys [49]–[51], but tools and practical systems have not been brought forward to our knowledge.

D. Summary

A methodology for quantifying the impact of entity integrity faults on database operations not only addresses the core for a rich research landscape on data integrity, data quality, data repairs and consistent query answering, but provides motivation for investment in further research and development to avoid or repair such faults, compute consistent answers, and build systems that facilitate the ultimate goals of high quality analytical insight and data-driven decision making.

III. Methodology for Impact Assessment

The main focus of our work is to provide a methodology that quantifies the impact of entity integrity faults on query accuracy. Figure 1 illustrates our proposal which we will describe now in detail.

Our methodology is based on several inputs. Firstly, we choose the dataset that will be investigated. In our case we have chosen the TPC-H benchmark for analysis. Secondly, we specify a workload that will be representative for what can be anticipated at the time. Results from analysing query logs could be harnessed to identify such a workload, for example. In our work we use the 22 TPC-H benchmark queries. Thirdly, we need to decide which dimensions of entity integrity faults we want to evaluate. These are determined by the different

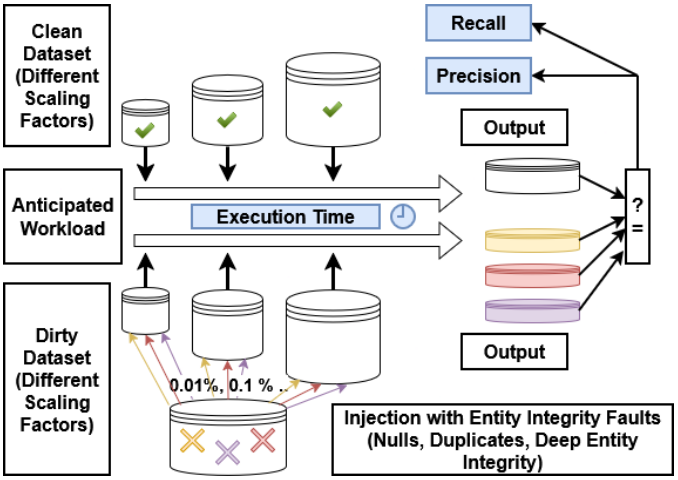

Fig. 1: Illustration of Methodology

causes of such faults. Here, we focus on the following three causes. Firstly, we investigate how *duplicate records* affect query accuracy. Secondly, we analyse the effect of *missing values* on primary key attributes. These two cases cover traditional problems related to entity integrity that primary key enforcement aims to alleviate. The third cause acknowledges the fact that surrogate keys alone cannot guarantee entity integrity. For that reason, we also measure the impact of what we call *deep entity integrity violations* on query accuracy. Here, real-world entities occur multiple times in the database because different artificial identifiers are used to represent them. All three causes are common cases of malpractice [4], [5], [52], and represent inhibitors to the popular data quality dimensions of uniqueness, completeness, and inconsistency, respectively. For each of the three causes, we choose different degrees of severity based on a relative amount of violations compared to the database size. Moreover, we aim to pinpoint which data source requires particular focus during a possible integration process to ensure entity integrity. In our case study we highlight in which tables of the database such violations occur for this purpose. We have chosen such violations at random and enterprises can adjust this based on available information about certain data sources and common malpractices in this regard. Finally, we also analyse how much the underlying database size matters for our analysis. In our case, this is naturally covered by the different scaling factors that apply to the TPC-H benchmark. Real-world datasets might need artificial scaling to achieve this or subsets of the investigated data repository might represent smaller scaling factors.

TABLE II: Answers on Original and Dirty Datasets

(a) Original Answers

orderpriority	order_count
1-URGENT	10781
2-HIGH	10484
3-MEDIUM	10366
4-NOT SPECIFIED	10587
5-LOW	10781

(b) New Answers

orderpriority	order_count
1-URGENT	11236
2-HIGH	10484
3-MEDIUM	10366
4-NOT SPECIFIED	11451

We have discussed the fundamental setting of our methodology. Now, we introduce the metrics to measure query accuracy. We run the dedicated workload on the original dataset and the dirty dataset that exhibits entity integrity violations. For each query we capture the output in both cases and compute recall and precision of the query. *Recall* measures how many of the clean query answers have been preserved. More precisely, recall takes the number of records that that the query answers on the clean and dirty dataset share and divides it by the number of query answers on the clean dataset. *Precision* measures how many of the new query answers are actually clean. More specifically, precision takes the number of records that are part of the query answers on the clean and dirty dataset and divides it by the number of query answers on the dirty dataset. An example is given by Table II where the records in yellow are both query answers on the original and dirty datasets. As there are 5 original answers, recall is 0.4 or 2/5; and since the new query has 4 answers, precision is 0.5 or 2/4.

Another important measure for database operations is performance. Hence, we also measure the *execution time* by which queries are evaluated. Since candidate keys, in particular the primary key, come with `unique` indices, query answering on the original dataset is supported by these structures. While speed-ups of query processing are to be expected through introduction of an index structure our methodology quantifies these benefits and pinpoints which indices are crucial and which are potentially negligible based on the provided workload. With regards to performance a dedicated database system can be chosen in the decision process when applying this framework while results regarding accuracy will be independent of this choice. Combining these results along with information on which queries are paramount to produce accurate answers can be used to guide the effort dedicated to enforce entity integrity and in which parts of the database this is most important.

IV. Experiments

Our experiments provide a proof of concept for our methodology. This is accomplished by quantifying the effects of entity integrity faults on the TPC-H workload. More precisely, we analyse how much violations of a primary key, such as duplicate records or occurrences of null, influence the accuracy of queries. For these scenarios we also measure potential performance gains that can be made through index structures that result from the presence of primary keys. Subsequently, we analyse deep entity integrity violations where entities are duplicated by assigning different primary key values. That is, primary keys are valid while some other business keys are violated. These experiment results are then used in a case study for decision support on entity integrity enforcement.

A. Dataset, Set Up and Measures

We use the popular TPC-H benchmark[2] for RDBMs. Its database schema is shown in Figure 2, reduced to primary

[2]https://www.tpc.org/tpch/

key (PK) and foreign key (FK) columns. For each table we see how the scaling factor (SF) determines how many records are generated. We used instances for three different scaling factors: small (0.01), medium (0.1) and large (1).

TABLE III: Hardware Specifications for Experiments

Component	Details
CPU	Intel(R) Core(TM) i7-10610U CPU 2.30 GHz
Cores	4 (8 threads)
Memory	64 GB (DD4 - 3200 SODIMM)
Disk	2 TB (PCIe(R) Gen4 NVMe(TM) 2280 M.2 SSD)
System type	64-bit

For each scenario in our experiments, every query has been run 100 times on each of the small and medium instance, and 50 times on the large instance due to constraints in the availability of computational resources. In each run, the time to execute the query has been recorded as well as its recall and precision. We refer the reader to the Github repository under https://github.com/GraphDatabaseExperiments/entity_integrity_faults_impact for further details on the experiments and artifacts. We used MySQL 8.0.40 along with Python 3.12.2. For the hardware specifications we refer to Table III.

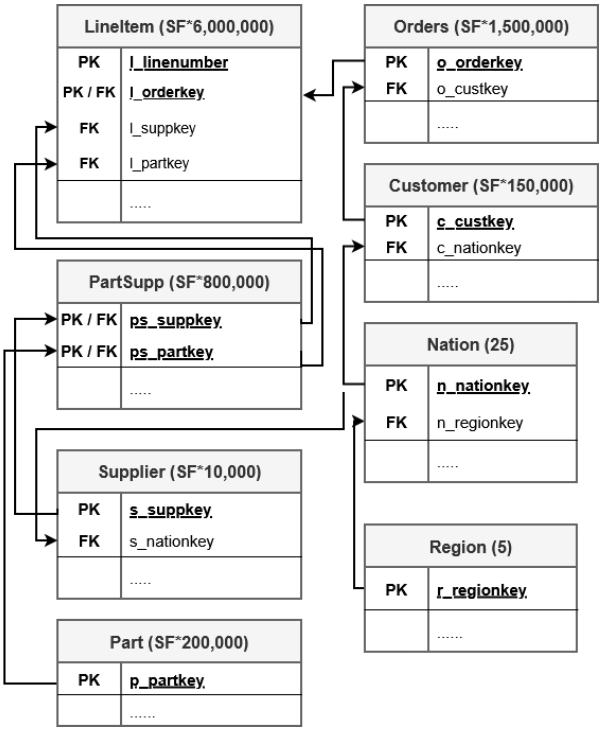

Fig. 2: Database Schema of the TPC-H Benchmark

B. Performance

We measure the performance gains achieved by `unique` indices associated with the primary key on each table. Such an index will automatically be created for every primary key and `unique` constraint. We measured the average execution time of each query run on the dataset, with and without index. From these results we recorded the improvement in execution time. Figure 3 shows the performance gains (in percent) for each query, and for each of the three scaling factors. These results show the improvements achieved on average. More precisely, for each table of TPC-H we run queries each time with and without the primary key index. This gives us the opportunity to pinpoint which indices are crucial and which structures do not significantly contribute to performance grains. The data points in Figure 3 show the average across all of these runs.

It is evident that the gains in performance grow with the size of the instance. Indeed, queries take uniformly more processing time without the primary key index. Unsurprisingly, how much performance is gained by the index depends on the query. Q1, for example, has barely any gain. For Q2, however, performance gains range from noticeable gains on the small instance to over 40% gains on the medium instance and around 80% on the large instance. This means that, on average, the presence of a single index reduces execution time to one fifth of what is required without it.

Execution times fluctuate between different runs. For this reason we focus on the large instance since fluctuations have only marginal impact on the overall execution time. Large performance gains can be observed for queries Q2, Q5, Q7, Q9 and Q21 while moderate gains can be noted for Q3, Q8, Q10, Q12, Q14 and Q18.

To get a better idea on which table an index is particularly relevant for a query we have detailed results in Table IV. Here, we have listed the performance gains from a particular index (Table) for each query for all scaling factors (SF). To only list noticeable performance gains we limited results to a change in query execution time of at least 5%. Across all scenarios we notice how benefits increase with the size of the instance. Performance gains for queries Q2, Q5, Q7 and Q21 are driven by an index on the *supplier* table. For Q2 we notice that heavy computational effort is required for joins of the *partsupp* table with *part* and *supplier* to determine the cheapest supplier within a region for certain parts. Here, the index on *partkey* in *part* and the one on *suppkey* in *supplier* are prime candidates to enhance query execution. The *supplier* table is larger than the *part* table and the effect becomes more pronounced there. In contrast, there are no significant speed-ups from the index on the *part* table. Unlike Q2, the queries Q5, Q7 and Q21 do not depend on the *part* table at all. Further investigation of Q9, where for each order year all qualifying records in *orders* must be joined with *lineitem*, shows great performance gains resulting from an index on *orders*.

For queries with moderate gains we see that Q3 benefits from an index on *lineitem*. For Q8 this is the case for *nationkey* in *nation*. Equally, this holds for Q10 and in addition index support on *customer* results in noticeable improvements there. For Q12 we notice that *orders* plays a crucial role in this regard and in the case of Q14 it is the *part* table that can provide moderate speed-up. For Q18 this seems more balanced and the benefits come from the index structure on *customer* and *lineitem*.

The remaining queries show only little improvement in

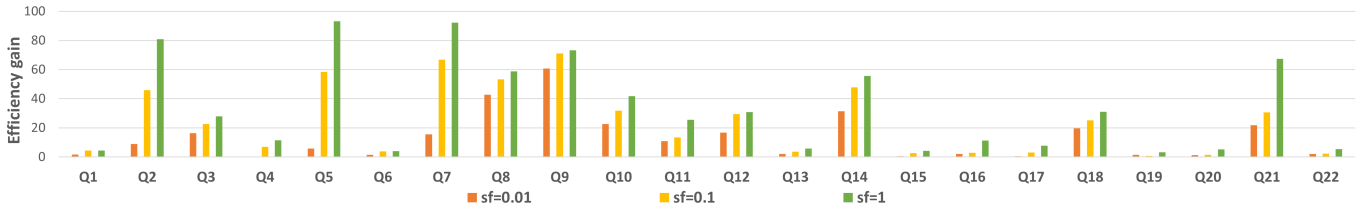

Fig. 3: Efficiency Gains for Queries through Primary Key and Associated Index Structure

TABLE IV: Effect of Index on Query Execution Speed of TPC-H Workload

Table	SF	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
region	0.01																						
	0.1																						
	1.0																						
nation	0.01		36.2%						86.9%	40.1%	33.1%												
	0.1		66.1%						90.3%	52.8%	47.7%												
	1.0		72.5%						92.1%	55.1%	58.3%												
customer	0.01										36.2%								18.9%				
	0.1										45.1%			6.4%					28.4%				
	1.0										57.7%			8.9%					35.0%				
orders	0.01			7.2%						88.7%			29.6%										
	0.1			13.3%				9.0%		92.5%			45.1%										
	1.0			18.5%				15.2%		93.2%			45.6%										6.7%
supplier	0.01					26.6%		37.9%		39.8%		10.1%										49.7%	
	0.1		69.6%			89.3%		90.4%		53.6%		12.1%				6.5%					6.2%	59.9%	
	1.0		94.7%			98.7%		98.2%		54.1%		12.6%										88.7%	
part	0.01														45.5%								
	0.1														64.9%								
	1.0														71.6%								
partsupp	0.01											17.6%											
	0.1											25.3%					7.0%						
	1.0		9.0%							6.2%		46.0%					19.9%						
lineitem	0.01			35.6%		11.6%		25.0%	7.8%		9.9%								33.9%			15.7%	
	0.1			43.7%	11.6%	17.4%		35.3%	11.2%		9.9%							8.7%	36.7%		8.4%	16.7%	
	1.0			48.2%	18.1%	23.5%		42.7%	12.6%		10.2%							16.0%	43.4%		11.7%	19.5%	

execution time when introducing indices on primary keys. In particular, for Q1 and Q6 the introduction of an index on the primary key of *lineitem* does not have an effect at all. Both times the query only makes use of this table and only non-key attributes are filtered and used for aggregation. The other queries showing little performance gains are less join-heavy and look-ups on the primary keys do not occur or are limited.

In summary, introducing primary keys and automatically generated indices significantly improves query execution time. While this is expected, our experiments show that not all queries benefit equally and that performance gains are often driven by specific tables. Notably, an index on the primary key of the largest table, *lineitem*, yields only modest improvements, whereas a similar index on the smaller *supplier* table provides substantially greater benefits. An index on *region* offers no measurable gain, which is unsurprising given its constant size of five records. In contrast, indexing *nation*, which consistently contains 25 tuples, results in considerable speed-ups for certain queries, likely because *nation* participates in multiple joins within the TPC-H workload. Our methodology thus indicates that maintaining an index on *nation*, which is relatively inexpensive, can deliver greater performance benefits for the observed workload than more costly indices on *part* or *partsupp*. Improved performance is a welcome side effect of enforcing entity integrity. In the following, we examine the impact of entity integrity faults on query accuracy.

C. Accuracy

Our experiments aim at quantifying the loss in accuracy for analytical queries when different degrees of entity integrity violations occur. For this purpose, we investigate three kinds of violations.

1) Record Duplication: First, we investigate the impact of duplicates on query accuracy. This means we duplicate records in a table, implying that the primary key for this table is turned off. For an example, we refer again to Table I where two city records are identical. In our experimental setup we have only introduced duplicates for one table at a time. This means we can easily locate from which source any impact of such violations on queries originates. In practice, this can be tailored to different sources of data acquisition.

The impact of duplicates on the TCH-H workload is summarized in Table V. For each table we have noted the percentage (0.01%, 0.1%, 1% and 10%) of records that we randomly duplicated. In case of the *region* and *nation* table we have introduced exactly one duplicate each time. Unlike the other tables, these two do not scale with the size of instances and always carry 5 and 25 records, respectively. It is worth noting that for the small instance and 0.01% duplication, the tables *customer*, *supplier*, *part* and *partsupp* are not affected as the percentage represents less than one record. The same applies to *supplier* for the small instance and 0.1% duplication, as well as for the medium dataset and 0.01% of duplicated entities. For each scaling factor (SF), Table V shows the mean and median value for recall (R) and precision (P) across all benchmark queries for different duplication factors.

We observe that a growing percentage of duplicates causes a drastic decline of query accuracy. What seems interesting is that for few entity integrity violations the median sits above the mean, which indicates that most queries provide the correct

result and some outliers related to incorrect outputs bring down the average. However, for larger instances this trend seems to reverse with a higher number of duplicates. This means that highly accurate outputs become the exception. Furthermore, for the same number of violations, accuracy declines with growing scaling factor. This effect may be due to the fact that some parameters, such as the date range of order placements, remain the same while the instance becomes larger. So, with growing size of the instance and fixed duplication factor there is a higher likelihood to have a duplicated order within a certain date range, which might distort the query output. Similar effects may be observed with regards to violations in the dataset elsewhere.

TABLE V: Impact of Duplicates on TPC-H Workload

		Percentage of Violations							
SF	mean/ median	0.01% R	0.01% P	0.1% R	0.1% P	1% R	1% P	10% R	10% P
0.01	mean	0.962	0.936	0.927	0.903	0.841	0.804	0.654	0.621
	median	1	1	1	1	1	1	0.883	0.8
0.1	mean	0.917	0.903	0.84	0.825	0.657	0.64	0.472	0.449
	median	1	1	1	1	0.963	0.96	0.404	0.277
1	mean	0.847	0.834	0.67	0.658	0.521	0.503	0.455	0.434
	median	1	1	0.996	0.992	0.908	0.8	0.358	0.05

Having provided a high-level overview, we now aim for a more detailed analysis. Going forward, we will fix the scaling factor of the dataset (sf = 1) and we keep the number of duplicates in corresponding tables at 0.1 percent. Now, we look at the effects of these integrity faults on recall and precision for each query. Figure 4 summarizes this scenario. Here, for each query, different sets of runs have been performed where, each time, one of the tables involved in the query contains duplicates and the others do not. This provides us with information on which queries are most affected while not yet focusing on the sources of data (here, which tables) that are contributing to these inaccuracies the most.

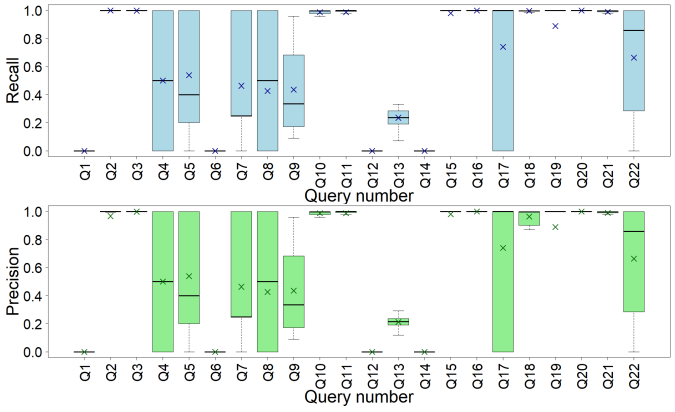

Fig. 4: Recall and Precision of Query Execution Grouped by Query Number with respect to Duplicates

Noticeably, recall and precision are often very similar and, in fact, coincide for several queries. Whenever they coincide the queries return the same number of answers on the original and dirty dataset. Whenever they differ the duplicates do not only cause a change in some values of the query answers, but leads to whole new answers or missing out on some previous answers. Noticeable deviation in recall and precision can be observed for Q13 and Q18, while minimal differences are present in the case of Q2 and Q11. For other queries such differences are marginal. In Table VI we summarized the average number of query answers on the original and dirty datasets. These numbers indicate that recall and precision do also not coincide for Q20. Again, these differences are marginal and seem to result from particular runs.

The TPC-H workload can be grouped into three general categories regarding the impact of duplicates on accuracy. Queries Q1, Q6, Q12 and Q14 are heavily affected, while Q2, Q3, Q10, Q11, Q15, Q16 and Q18 - Q21 still provide highly correct outputs with the exception of Q19 which suffers slightly more than others. The remaining queries lie somewhere in between, exhibiting varying degrees of accuracy loss. Based on which queries are deemed most crucial with regards to correctness decisions based around introduction of thorough entity integrity checks can be made.

Now that we have discussed the effect of duplicates grouped by queries we like to focus on the sources of these inconsistencies. This breakdown is illustrated in Figure 5. Here, we take a look at how duplicate records affect the TPC-H workload with respect to where in the database they occur. The different shading of the plots for the *region* and *nation* tables indicates that here, consistently one tuple has been duplicated, unlike 0.1 percent of tuples in the other tables.

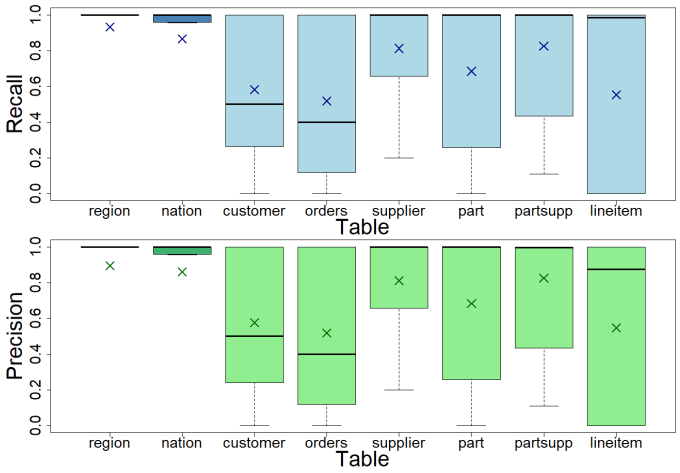

Fig. 5: Recall and Precision of Query Execution Grouped by Tables with respect to Duplicates

First, we observe no fundamental divergence between recall and precision overall. Notably, when duplicates are introduced in the *lineitem* table, median precision is lower than median recall, while the means of both measures align. Duplicates in *region* and *nation* have little impact, though the lower mean values suggest that a few affected queries suffer severe degradation. Duplicates in other tables influence TPC-H queries to varying degrees, with *customer* and *orders* showing particularly low mean and median recall and precision. This indicates that avoiding duplicates in these tables is especially

TABLE VI: Average Size of Output from TPC-H Workload on Original and Dirty Dataset (with Duplicates)

Dataset	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
original	4	471.24	11428.89	5	5	1	3.84	2	175	37737.56	890.69	2	42	1	1	18266.69	1	803.23	1	173.65	375.38	7
dirty	4	497.61	11428.89	5	5	1	3.84	2	175	37737.56	891.03	2	46.16	1	1	18266.69	1	833.07	1	173.69	375.38	7

TABLE VII: Effect of Duplicates on Accuracy of TPC-H Workload with sf=1

Table	%D	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
region			1.0			0.8			1.0														
nation			1.0			0.948		1.0	0.04	0.96	0.96	0.952									1.0	0.978	
customer	0.01			0.999		0.912		0.866	0.91		0.999			0.702					0.999				0.877
	0.1			0.999		0.3		0.308	0.43		0.999			0.281					0.999				0.328
	1.0			0.989		0.0		0.0	0.02		0.99			0.147					0.99				0.0
	10.0			0.9		0.0		0.0	0.0		0.899			0.072					0.9				0.0
orders	0.01			0.999	0.356	0.912		0.883	0.84	0.827	0.999		0.21	0.393					0.999			0.999	1.0
	0.1			0.999	0.0	0.264		0.25	0.37	0.159	0.998		0.0	0.19					0.999			0.99	1.0
	1.0			0.99	0.0	0.0		0.0	0.0	0.0	0.987		0.0	0.07					0.989			0.906	1.0
	10.0			0.9	0.0	0.0		0.0	0.04	0.0	0.874		0.0	0.026					0.901			0.37	1.0
supplier	0.01		1.0			0.948		0.933	0.93	0.962		0.998				1.0					1.0	0.999	
	0.1		1.0			0.664		0.635	0.37	0.686		0.992				1.0					1.0	0.999	
	1.0		1.0			0.016		0.026	0.1	0.02		0.924				1.0					1.0	0.99	
	10.0		1.0			0.0		0.0	0.04	0.0		0.436				1.0					1.0	0.898	
part	0.01		1.0						0.92	0.894					0.0		1.0	1.0		1.0	1.0		
	0.1		1.0						0.41	0.329					0.0		1.0	0.86		0.9	1.0		
	1.0		1.0						0.0	0.0					0.0		1.0	0.18		0.34	1.0		
	10.0		1.0						0.04	0.0					0.0		1.0	0.1		0.0	1.0		
partsupp	0.01		1.0							0.888		0.999					1.0						
	0.1		1.0							0.326		0.992					1.0						
	1.0		1.0							0.0		0.921					1.0						
	10.0		1.0							0.0		0.419					1.0						
lineitem	0.01	0.015		0.999	1.0	0.848	0.0	0.893	0.94	0.824	0.999		0.19		0.0	1.0		0.94	0.999	0.98	1.0	0.999	
	0.1	0.0		0.997	1.0	0.236	0.0	0.245	0.34	0.161	0.997		0.0		0.0	0.92		0.62	0.993	0.88	1.0	0.989	
	1.0	0.0		0.973	1.0	0.0	0.0	0.0	0.0	0.0	0.97		0.0		0.0	0.34		0.12	0.933	0.18	1.0	0.9	
	10.0	0.0		0.767	1.0	0.0	0.0	0.0	0.04	0.0	0.742		0.0		0.0	0.0		0.22	0.479	0.0	0.999	0.369	

critical. Across all tables except *region* and *nation*, recall and precision exhibit substantial variability, highlighting that the absence of entity integrity can lead to highly inaccurate results. A likely explanation is that many TPC-H queries aggregate over specific nations and regions and thus remain correct despite duplicates in those tables depending on parameter choice. In contrast, queries that count customers or orders per customer under certain conditions are highly sensitive to duplicates in the *customer* and *orders* tables.

Overall, duplicates in tables affect queries that use that table to different degrees. Table VII quantifies how much certain percentages of duplicates in specific tables impact the recall of queries on average. This outlines to which level correct outputs are missing. A similar heat map can be generated showing query precision which we left out due to space restrictions. As mentioned before, in most cases precision and recall are only marginally different, if at all.

Table VII shows that the recall of a query is affected quite dramatically by only modest percentages of duplication (%D). In the following, we investigate this effect in a more nuanced manner. The majority of queries use aggregation. Q1 and Q6 are affected significantly by duplication in *lineitem*, which is the only table involved in these queries. Q12 and Q14 exhibit similar behaviour across multiple tables that affect them. A slightly different picture emerges for Q4, Q8, Q15 and Q22. For Q8, duplicates in *region* have no effect at all. Indeed, the query filters by parts that satisfy some conditions and have been ordered by customers from some nation in some region. Hence, joining the *nation* and *region* table carrying some duplicated record will not yield a different result. This behaviour is mirrored by Q4 and Q22, yet there it is the *lineitem* and *orders* table respectively, that does not impact query accuracy. In both queries existence checks are conducted on the corresponding table, hence duplicates have no effect on the output. Q15 is impacted by duplicates in *lineitem*, but not by violations in *supplier*. The situation for Q17 and Q19 is comparable to Q12 and Q14, only that accuracy is not affected as heavily for the latter. Queries Q3, Q5, Q7, Q9, Q10, Q11, Q13, Q16, Q18 and Q21 show various degrees of decline in accuracy, depending on which table carries duplicates. Query Q16 remains unaffected as aggregation occurs over distinct records. For Q20 we notice that only duplicates in *lineitem* can impact this query, and even then a different output is highly unlikely. The query aggregates the quantity of *lineitem* records associated with particular parts and grouped by suppliers who supplied this part. Then the query checks if the supplier of these parts has enough capacity. As the difference between this capacity and the aggregated values from *lineitem* is quite substantial, most of the time even high levels of duplication in *lineitem* can only rarely distort results. On top of that, an additional filter that looks for suppliers in certain countries makes a change in output even less likely. Finally, we analyse the only truly pointy query, Q2. Duplicates in any table might either not affect query Q2 at all or if they do, some suppliers (along with the corresponding nation name and part columns) will be recorded multiple times. However, the original output will always be retained by the new output. Hence, recall is never impacted. Table VI provides further insight, and also explains the difference between recall and precision for Q2.

In summary, our results show that duplicate records, as a form of entity integrity violation, can severely degrade query accuracy. Thankfully, duplicates are easy to detect and fix. Our experiments identify which tables must be kept duplicate-free to preserve accuracy for TPC-H queries. Surprisingly, duplication in the *region* and *nation* tables has limited impact even though we would expect the effects of duplicates to be

strongly propagated to child tables. In contrast, other tables show significantly more impact. The effects of entity integrity violations is multifaceted, with each query exhibiting a distinct sensitivity to duplication, influenced also by query parameters. Notably, even relatively low levels of duplication can cause substantial inaccuracies, including missing previous results or introducing new ones. Moreover, for a fixed duplication rate, accuracy declines further as database size increases.

Based on these findings, our methodology can guide decisions and resource allocation. For example, ensuring the absence of duplicates in *partsupp* may be less critical than in other tables, particularly if the accuracy of Q9 and Q11 is less important and considering that indexing *partsupp* yields only limited performance benefits.

2) Null Violation: In this experiment, primary keys were turned off for those TPC-H tables where we introduced null markers in some columns of the key. Table I shows a city record in which the ”zip” column is NULL.

TABLE VIII: Impact of Null Violations on TPC-H Workload

		Percentage of Violations							
SF	mean/ median	0.01% R	0.01% P	0.1% R	0.1% P	1% R	1% P	10% R	10% P
0.01	mean	0.971	0.951	0.954	0.933	0.893	0.874	0.692	0.695
	median	1.000	1.000	1.000	1.000	1.000	1.000	0.889	1.000
0.1	mean	0.944	0.936	0.884	0.875	0.701	0.695	0.532	0.542
	median	1.000	1.000	1.000	1.000	0.978	0.998	0.826	0.842
1	mean	0.882	0.873	0.711	0.704	0.581	0.576	0.518	0.529
	median	1.000	1.000	0.997	0.999	0.961	0.982	0.836	0.834

First we show how missing primary key values impact precision and recall of the queries in TPC-H. A summary is shown in Table VIII. Again, we have performed experiments with different percentages of integrity faults in each table, and for *region* and *nation* we consistently introduced one record with null violations. The results in Table VIII are similar to those in Table V. Comparing these insights, we notice that duplicates have a larger impact than nulls. This is particularly noticeable for higher error levels.

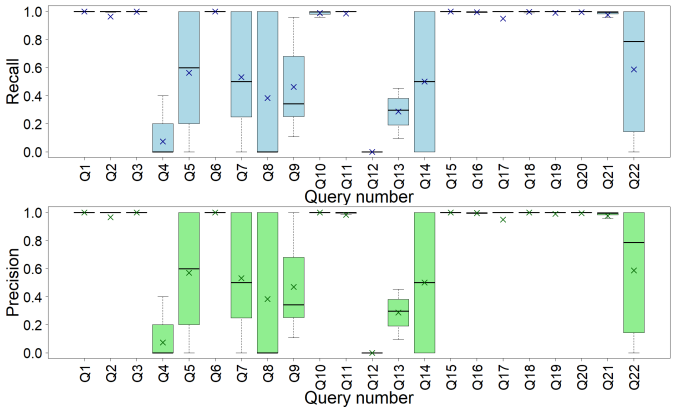

Fig. 6: Recall and Precision of Query Execution Grouped by Query Number with respect to Null Violations

Like in previous experiments we fix the TPC-H scaling factor to 1, and the percentage of null violations in relevant tables to 0.1%. Figure 6 shows a breakdown of recall and precision for each query with respect to different tables that exhibit integrity faults. Figure 6 outlines how the workload is affected by missing values on primary key attributes in general. Only marginal differences between recall and precision for each query are visible, with Q10 displaying the largest difference. Table IX shows the amount of records in the output of each query on average which provides further insight regarding the two metrics. The results resemble those shown in Table IX. However, we can infer that query recall and precision differ more under null violations than in the presence of duplicates. Most notably, Table IX shows that missing primary key values often lead to queries with fewer answers than they should. Hence, there is a decrease in recall. For most queries this will be due to joins not being performed. This represents a major difference from how duplicates affect queries.

In the following we investigate the effect of null violations with respect to the tables that exhibit these integrity issues. This reflects how different data sources during the data integration process might affect accuracy. Figure 7 shows only marginal differences between precision and recall. In this regard the effects of violations caused by duplicates are similar to those that result from nulls.

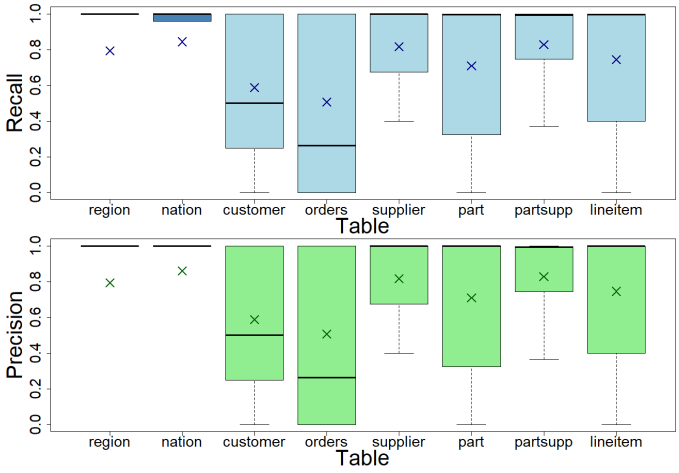

Fig. 7: Recall and Precision of Query Execution Grouped by Tables with respect to Null Violations

Now we provide a detailed breakdown of how each query is affected by the percentage of null violations (%N) and how these violations in relevant tables impact recall of query answers. The results are summarized in Table 10. Compared to Table 7, duplicates and null violations have similar effects on query accuracy in terms of which tables cause inaccuracies and their severity. A key difference is the impact of null violations in the *lineitem* table: queries Q1, Q6, Q14, Q17, and Q19 are unaffected by null violations, whereas duplicates do affect them. This is likely because these queries do not join *lineitem* with *orders* or otherwise rely on its primary key. Conversely, Q4 shows the opposite behavior: null violations in *lineitem* affect recall due to the join with *orders*, while duplicates do not, since the query filters orders based on conditions over *lineitem*, making duplicated *lineitem* records irrelevant. Additionally,

TABLE IX: Average Size of Output from TPC-H Workload on Original and Dirty Dataset (with Null Violations)

Dataset	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
original	4	472.05	11451.65	5	5	1	3.66	2	175	37667.225	922.32	2	42	1	1	18273.25	1	793.48	1	170.77	377.99	7
dirty	4	455.18	11443.16	5	4.74	1	3.57	1.94	173.83	37271.14	912.23	2	42	1	1	18268.07	1	791.67	1	169.80	371.68	7

TABLE X: Effect of Null Violations on Accuracy of TPC-H Workload with sf=1

Table	%N	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
region			0.738			0.76			0.74														
nation			0.917			0.94		0.976	0.02	0.96	0.959	0.968									0.941	0.958	
customer	0.01			0.999		0.9		0.907	0.92		0.999			0.791					0.999				0.845
	0.1			0.999		0.368		0.391	0.37		0.999			0.381					0.998				0.174
	1.0			0.99		0.0		0.0	0.06		0.99			0.196					0.99				0.0
	10.0			0.9		0.0		0.0	0.06		0.899			0.085					0.899				0.0
orders	0.01			0.999	0.32	0.844		0.843	0.95	0.829	0.999		0.25	0.392					0.999			0.998	1.0
	0.1			0.999	0.0	0.232		0.232	0.26	0.169	0.998		0.0	0.193					0.999			0.99	1.0
	1.0			0.99	0.0	0.0		0.0	0.04	0.0	0.987		0.0	0.064					0.989			0.908	1.0
	10.0			0.899	0.0	0.0		0.0	0.02	0.0	0.874		0.0	0.002					0.9			0.369	1.0
supplier	0.01		0.999			0.936		0.949	0.94	0.963		1.0				1.0					0.999	0.999	
	0.1		0.998			0.672		0.744	0.29	0.68		0.998				1.0					0.999	0.998	
	1.0		0.99			0.02		0.01	0.08	0.016		0.987				0.98					0.99	0.99	
	10.0		0.902			0.0		0.0	0.02	0.0		0.869				0.84					0.9	0.901	
part	0.01		1.0						0.86	0.887					0.02		0.999	1.0		0.98	0.999		
	0.1		0.999						0.48	0.347					0.0		0.998	0.9		0.98	0.999		
	1.0		0.99						0.04	0.0					0.0		0.983	0.48		0.22	0.992		
	10.0		0.902						0.0	0.0					0.0		0.845	0.2		0.0	0.922		
partsupp	0.01		0.999							0.889		1.0					0.999						
	0.1		0.998							0.326		0.998					0.993						
	1.0		0.99							0.0		0.985					0.937						
	10.0		0.904							0.0		0.867					0.531						
lineitem	0.01	1.0		0.999	0.844	0.904	1.0	0.875	0.91	0.88	0.999		0.34		1.0	1.0		1.0	0.999	1.0	1.0	0.997	
	0.1	1.0		0.998	0.148	0.416	1.0	0.393	0.46	0.289	0.998		0.0		1.0	1.0		1.0	0.995	1.0	1.0	0.976	
	1.0	1.0		0.982	0.0	0.0	1.0	0.0	0.08	0.0	0.98		0.0		1.0	1.0		1.0	0.957	1.0	1.0	0.791	
	10.0	1.0		0.84	0.0	0.0	1.0	0.0	0.04	0.0	0.822		0.0		1.0	1.0		1.0	0.621	1.0	1.0	0.255	

null violations in any table involved in Q2 affect recall, unlike in the case of duplicates, likely because some joins cannot be executed correctly. Similar behavior is observed for Q16. Query Q8 is also affected by null violations in the *region* table, as missing values prevent correctly associating nations with regions; duplicates in this table have no effect. Queries Q7 and Q20 show similar behavior with respect to *nation* and *region*. Finally, unlike duplicates in the *supplier* table, null violations affect joins between *lineitem* and *supplier* in Q15, as well as in Q16 and Q20 with their respective tables.

Overall, like duplicated records, missing primary key values can severely impact query accuracy, often resulting in missing or incorrect answers. While this is expected, the impact of null violations is not more severe than that of duplicates. Notably, missing primary key values affect joins in cases where duplicates do not, which might suggest that avoiding such errors is more critical than preventing duplication. However, this does not appear to be the case. Conversely, aggregate queries that do not involve primary key columns are unaffected by null violations. Decisions about where to allocate resources to prevent null violations should depend on which queries require the highest accuracy and which data sources are most prone to such errors. One possible approach is mining business keys and using them as primary keys, or introducing artificial identifiers based on these keys, while also considering trade-offs such as performance gains and index maintenance costs.

3) Deep Entity Integrity Violation: In this part of our experimental study, we examine the impact of deep entity integrity violations due to the presence of semantic duplicates in the data. In general, this is possible because some business key is violated.

Table I shows an example of multiple customer records with different identifiers. Despite satisfying artificial identifier constraints, other meaningful keys may be violated, highlighting the need for business keys based on natural attributes for entities. Our experiments involve the duplication of entities with different identifiers in one table, and random reassignment of references to either identifier from records in the child tables. We refer to such a scenario using child and parent table names connected through an arrow, for example *customer* ← *orders*. The sole use of surrogate keys has shortcomings as our experiments highlight.

TABLE XI: Impact of Deep Entity Integrity Violations on TPC-H Workload

		Percentage of Violations							
SF	mean/ median	0.01% R	0.01% P	0.1% R	0.1% P	1% R	1% P	10% R	10% P
0.01	mean	0.994	0.985	0.977	0.962	0.943	0.922	0.863	0.842
	median	1	1	1	1	1	1	1	1
0.1	mean	0.98	0.974	0.952	0.938	0.892	0.882	0.83	0.822
	median	1	1	1	1	1	1	1	1
1	mean	0.956	0.944	0.9	0.891	0.856	0.844	0.815	0.809
	median	1	1	1	1	1	1	1	1

We have conducted experiments for violations *orders* ← *lineitem*, *customer* ← *orders* and the scenarios *part* ← *partsupp* ← *lineitem* and *supplier* ← *partsupp* ← *lineitem*. In the last two cases we have propagated changes of the primary key value up to the *lineitem* table. These four situations present realistic data integrity issues. They can occur when for example, the same customer is stored with multiple customer ids. This can happen when solely using surrogate identifiers. Naturally, what follows is that some orders are associated with one representation of the customer, and other orders are attributed to another representation.

The effect of deep entity integrity violations on the TPC-H workload for different scaling factors is outlined in Table XI. While dataset size and violation frequency influence accuracy,

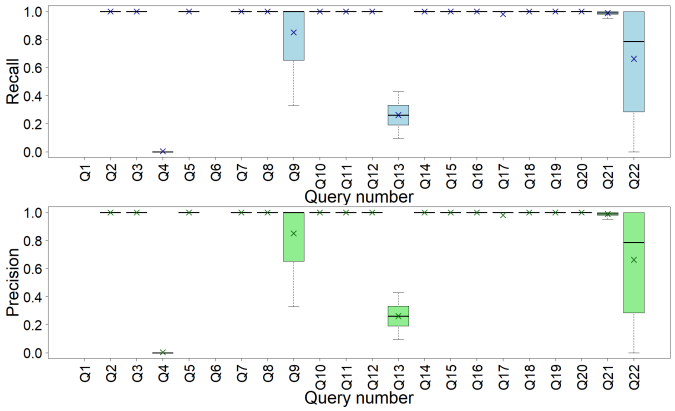

Fig. 8: Recall and Precision of Query Execution Grouped by Query Number with respect to Deep Entity Integrity Violations

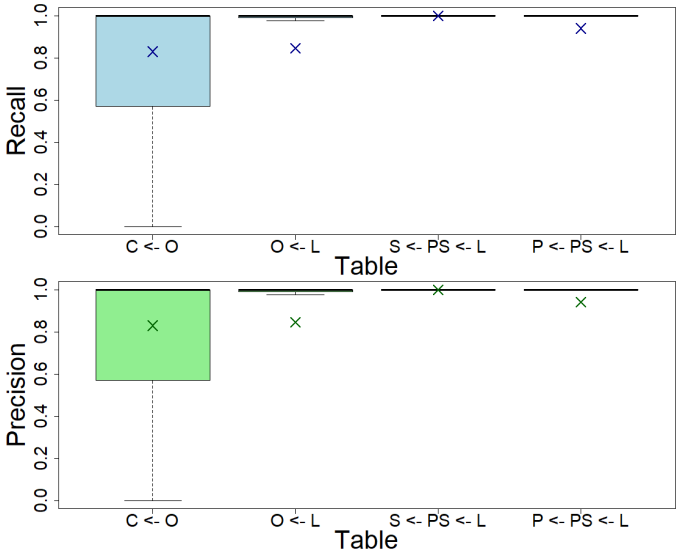

Fig. 9: Recall and Precision of Query Execution Grouped by Tables with respect to Deep Integrity Violations

deep entity integrity violations generally cause slower degradation compared to duplicates or null violations. For a more nuanced breakdown we have again fixed the scaling factor sf at 1 and considered deep entity integrity violations for 0.1%.

Figure 8 shows a breakdown of the impact by query. Results for Q1 and Q6 are excluded as none of the scenarios for which we conducted experiments affect these. Q4 is heavily affected, followed by Q13, Q22, Q9 and Q21. The dirty dataset does either not affect the remaining queries at all or only marginally. Under deep entity integrity violations, recall and precision seem to align more than in the previously discussed scenarios. This is supported by the results shown in Table XII. We note that the 'x' indicates that no outputs were recorded as the queries had not been affected.

We have further investigated the impact of violations based on the table they originate from (e.g., *customer* or *orders*). The results are shown in Figure 9. The most significant effects originate from *customer* and *orders* errors, with some outliers affecting query accuracy more in the latter. Violations in the *supplier* and *part* table, however, have less impact.

Table XIII summarizes the impact of varying levels of deep integrity violations (%V). The impact from *supplier* ← *partsupp* violations is minimal, while the presence of semantic duplicates in other scenarios affect specific queries to varying degrees. These include Q3, Q4, Q9, Q10, Q13, Q17, Q18, and Q20-Q22. The remaining queries fall into two categories. Some queries, like Q1, are unaffected. That is, *orders* ← *lineitem* does not impact foreign key columns in *lineitem*, the only table that Q1 uses. Others, such as Q2, remain unchanged because the violations do not alter the results of the query. Selections and joins still yield identical outputs with and without deep integrity faults. Query Q13, however, suffers substantial accuracy loss under both, *orders* ← *lineitem* and *customer* ← *orders*, scenarios due to incorrect aggregation. Since the query aggregates lineitem values associated with orders per customer, either violation is likely to distort the results. In contrast, Q10 shows minimal accuracy loss, with aggregation over *lineitem* still correct, though *customer* ← *orders* causes incorrect outputs sometimes. Here, aggregation over *lineitem* is still executed correctly in the scenario *orders* ← *lineitem* as the query filters by *orders* and performs summation over associated *lineitem* records. Yet, the query occasionally produces incorrect outputs under the violation *customer* ← *orders* as primary keys of customers are returned. In this case, accuracy degradation correlates with the percentage of errors. Results for Q3, Q4, Q9, Q17, Q18, Q20-Q22 follow similar patterns to the previously discussed scenarios.

We conclude that deep entity integrity violations can significantly reduce query accuracy, highlighting the importance of enforcing business keys alongside artificial identifiers. While in our experiments the impact of semantic duplicates is generally less severe than the impact of duplicates or null violations, these errors may be more subtle and harder to detect and fix. The severity depends on the percentage of violations and database size, though the effect is more concentrated on specific queries, with recall and precision aligning better than in scenarios involving duplicates or missing primary key values. Our results for example show that dedicating resources to avoid deep integrity issues originating from the *supplier* might be less important than avoiding semantic duplicates in the *customer* table. This is particularly true if accuracy for either Q13 or Q22 is paramount. One way to detect these errors is through finding violations of meaningful keys. Once data is cleaned we can introduce business key constraints which are supported by an index structure that can lead to further query performance improvements. These aspects should all be taken into account when making decisions regarding entity integrity enforcement. For details on all conducted experiments we refer the reader to our Github repository.

After having analysed the accuracy of the proposed query workload on the provided dataset along with granular results regarding performance benefits introduced though index structures that accompany constraint creation we will show how

TABLE XII: Average size of output from TPC-H workload on original and dirty dataset (with deep entity integrity violations)

Dataset	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
original	x	469.13	11453.26	5	5	x	3.71	2	175	37575.69	977.38	2	42.00	1	1	18270.68	1	806.35	1	168.67	391.5	7
dirty	x	469.13	11455.80	5	5	x	3.71	2	175	37578.29	977.38	2	42.08	1	1	18270.68	1	805.99	1	168.66	391.5	7

TABLE XIII: Effect of deep entity integrity violations on accuracy of TPC-H workload with sf=1

Scenario	%V	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
$c \leftarrow o$	0.01			1.0		1.0		1.0	1.0		0.999			0.62					0.999				0.914
	0.1			1.0		1.0		1.0	1.0		0.999			0.327					0.999				0.325
	1.0			1.0		1.0		1.0	1.0		0.994			0.179					0.995				0.0
	10.0			1.0		1.0		1.0	1.0		0.943			0.08					0.948				0.0
$o \leftarrow l$	0.01			0.999	0.556	1.0		1.0	1.0	1.0	1.0		1.0	0.384					0.999			0.998	1.0
	0.1			0.999	0.004	1.0		1.0	1.0	1.0	1.0		1.0	0.197					0.999			0.978	1.0
	1.0			0.992	0.0	1.0		1.0	1.0	1.0	1.0		1.0	0.069					0.99			0.806	1.0
	10.0			0.925	0.0	1.0		1.0	1.0	1.0	1.0		1.0	0.031					0.902			0.185	1.0
$s \leftarrow ps \leftarrow l$	0.01		1.0			1.0		1.0	1.0	1.0		1.0				1.0					1.0	1.0	
	0.1		1.0			1.0		1.0	1.0	1.0		1.0				1.0					1.0	1.0	
	1.0		1.0			1.0		1.0	1.0	1.0		1.0				1.0					1.0	1.0	
	10.0		1.0			1.0		1.0	1.0	1.0		1.0				1.0					1.0	0.999	
$p \leftarrow ps \leftarrow l$	0.01		1.0						1.0	0.948					1.0		1.0	0.98		1.0	1.0		
	0.1		1.0						1.0	0.551					1.0		1.0	0.98		1.0	0.999		
	1.0		1.0						1.0	0.003					1.0		1.0	0.74		1.0	0.995		
	10.0		1.0						1.0	0.0					1.0		1.0	0.22		1.0	0.961		

our methodology can be applied to guide the decision making process when it comes to entity integrity enforcement.

4) Decision Support: We present a case study illustrating how our framework can support decisions on enforcing entity integrity. Suppose accurate results for query Q5 are business-critical. Enforcing entity integrity in the *region* and *nation* tables is straightforward due to their small size and can even be done manually. In contrast, the *supplier* table is more challenging. Here, both null violations and duplicates substantially affect query results. Based on business requirements, we can assess the impact of such errors on aggregated outputs. For example, with 0.1% duplicated *supplier* entries at scale factor 1, the revenue for two of the five countries returned by Q5 changes by 0.43% and 0.19%, respectively, illustrating error amplification. To put these numbers into perspective, in an extreme case, duplicating a single supplier (*suppkey* = 3020) alters the revenue for 'ROMANIA' by 1.13% and even changes the result ordering. Similar effects occur with null violations, whereas deep integrity violations have negligible impact in this part of the database. Given these findings, and the fact that enforcing a primary key on *supplier* yields significant performance gains for multiple queries, we conclude that a primary key on *supplier* should be enforced. However, additional effort to detect semantic duplicates to prevent deep integrity violations is unnecessary. Yet, this might be different for other tables. This approach can be applied to various parts of the database to guide integrity enforcement across the workload.

V. Conclusion and Future Work

We have proposed a methodology for assessing the impact of entity integrity violations on workload accuracy across multiple data quality dimensions. It helps database practitioners understand by how much different types of entity integrity violations affect queries, including duplicate records, partial records and deep entity integrity violations with respect to the sources they originate from. In addition, this framework highlights where index structures can offer the greatest performance improvements. Combining these aspects helps to form decisions around allocation of resources to guarantee entity integrity. Using the TPC-H benchmark, we demonstrated that even small entity integrity violations can cause significant inaccuracy in analytical query answers. Our findings stress the importance of robust schema design with regard to entity integrity that incorporates business keys alongside artificial identifiers. Our framework aids decision-making in this design process.

The methodology provides a first step towards measuring the impact of data quality issues on data workloads, offering several avenues for future investigation. One potential direction is the application of our methodology to other relational benchmarks, which could provide further showcases into the impact of entity integrity violations. Additionally, an analysis of real-world datasets that already exhibit these violations could yield valuable findings. However, as noted by the authors in [7], using real-world datasets presents challenges in identifying a ground truth, which necessitates detecting and eliminating all entity integrity violations beforehand. Another promising research direction could involve examining datasets based on different data models [13]–[15], [18], [19]. Extending our approach to incorporate additional data quality [32] and big data dimensions [53] would further enrich our framework. These extensions could address both extrinsic and generic scenarios, as well as intrinsic and tailored ones as discussed in [7]. Lastly, the investigation of other types of data integrity violations and applying our framework to these cases represents an interesting avenue for future research.

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